

CSS7000 - INTERNSHIP

Review- Presentation

GeoDefence: Interactive Threat Mapping and Monitoring System

Student Details	
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Under the Supervision of,

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**Name of the Program: B. Tech Computer
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Name of the School Internship Coordinator: Dr. Geetha Arjunan



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About Company or Organization

- ATWIC R&D is a deep-tech research and engineering company headquartered in Kakkanad, Kochi, Kerala.
- The company specializes in advanced sensing technologies, electronic systems, and integrated engineering solutions.
- Its solutions support applications in defense, aerospace, oceanography, environmental sciences, scientific research, and industry.
- Founded by Rahul Sivan, ATWIC R&D brings together multidisciplinary expertise in:
 - Hardware Design
 - Firmware Development
 - Mechanical Engineering
 - Scientific Research
 - Signal Processing
 - AI-Based Implementation
- The company combines scientific principles with practical engineering to transform complex physical phenomena into measurable, reliable, and usable technological solutions.



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Project Overview

- Interactive threat mapping and monitoring system.
- Mark and manage location-based observations.
- Classify threats as High, Medium, or Low.
- Supports Normal, Terrain, and Satellite map views.
- Includes a dashboard for monitoring detections.
- Developed using Python, PyQt6, Leaflet, HTML, CSS, and JavaScript.



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Working domain or the technology

- **Python** – Core programming and application logic.
- **PyQt6** – Desktop application and GUI development.
- **QWebEngineView** – Displays the interactive map within the application.
- **HTML, CSS & JavaScript** – User interface and interactive functionality.
- **Leaflet.js** – Interactive map visualization and marker management.
- **OpenStreetMap, OpenTopoMap & Esri World Imagery** – Normal, terrain, and satellite map views.



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About your team and reporting Manager

Team Members:

FELIX K S — 20241CAI0086

5th sem AI & ML Student

Presidency University , Bengaluru

Reporting Manager:

Mr.SREERAG

Senior AI/ML Developer

Reporting Manager at ATWIC R&D



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Challenges Faced in Internship

- Integrating multiple technologies into a single application.
- Implementing interactive map layers, markers, and user interactions.
- Managing different map views and tile services.
- Designing a user-friendly monitoring dashboard.
- Debugging and testing features such as marker movement, threat classification, and deletion.
- Converting project requirements into practical features.



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Objectives of the work

- Develop an interactive map for geographical monitoring.
- Enable users to mark location-based detections.
- Provide **Normal, Terrain, and Satellite** map views.
- Classify detections as **High, Medium, or Low** threats.
- Display detection information and statistics through a monitoring dashboard.
- Develop a standalone desktop application using **Python and PyQt6**.
- Gain practical experience in integrating Python with web and mapping technologies.



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Github Link

<https://github.com/felixsajeev2006-blip/GeoDefence-Interactive-Threat-Mapping-and-Monitoring-System>



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Q&A



Thank you !!



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